AP CHEMISTRY Sem-B : Syllabus 2026-2027
Teacher- Boppana; Room 247
Office Hours : First 20mins of Lunch M-F
Contact: jodi_boppana@mcpsmd.org

Daily Expectations

- **Punctuality & Readiness:** Arrive on time and fully prepared to learn.
- **Personal Mobile Devices (PMDs):** PMDs must be stored out of sight at all times during the school day.
- during the school day.

Grading & Reporting

- All assignment grades will be posted directly in **Canvas and synced to Synergy**
- Students are responsible for regularly monitoring their grades and promptly notifying Ms. Boppana if any discrepancies arise.

Reassessment Policy

- Students may reassess up to **two Assessment Tasks (ATs)** per quarter. PLC Designated
- *Note: Major Unit Assessments are not reassessable.*

Absences & Make-Up Work

- **Notification:** Please send an email to Ms. Boppana as soon as possible when you are absent.
- **Timeline:** Students will have the number of days missed plus one (not to exceed 5) to make up work. Students should arrange a make-up plan with Ms. Boppana right away—the sooner, the better!

Academic Integrity & AI Policy

- All submitted work must be strictly student-owned and adhere to the **Churchill Code of Conduct and Student Expectations**. Unapproved AI usage or plagiarism will result in disciplinary and academic review according to school policy.

Required Daily Materials

- Fully charged, school-issued **Chromebook**
- Writing utensil (**pen or pencil**)
- **AP Reference Packet**

AP Chemistry Syllabus 2026-2027
Semester B
Note: Quarter 3 starts as of 1/26
Note II: Summative assessments starting with Unit 11 include chapter content from the first semester, specific chapters are provided in the section for testing in the syllabus

*Unit 8 – Atomic Structure: Explaining the Periodic Properties of Elements
1/26 to 2/15

Non-instructional Days: 1/25 no classes, end of semester grading; 2/15 Holiday, President's Day

Read Chapter 3 and section 8.2, specifically the section on Determining Concentration (this covers the Beer – Lambert Law for determining concentration from light absorbance.)

Chapter 3: Atomic Structure: Explaining the Properties of Elements

3.1 The Electromagnetic Spectrum and the Stories It Tells Us
3.2 Atomic Spectra
3.3 Particles of Light: Quantum Theory
3.4 The Hydrogen Spectrum and the Bohr Model
3.5 Electrons as Waves
3.6 Quantum Numbers
3.7 The Sizes and Shapes of Atomic Orbitals
3.8 The Periodic Table and Filling Orbitals
3.9 Electron Configurations of Ions
3.10 The Sizes of Atoms and Ions
3.11 Ionization Energies and Photoelectron Spectroscopy (PES)
3.12 Electron Affinities (Not testable, good FYI material)

Read the Summary on page 135
Review the Problem-Solving Summary on page 136

Tuesday 02/16: AP Chem Exam #8, Chapter 3, Section 8.2, and Stoichiometry/reactions:Chapters 1, 2, 7 and 8

Unit 9 – Chemical Bonding, Bonding Theories and Molecular Geometry

2/17 to 2/26

Early release days: 2/26

Read Chapter 4: Chemical Bonding – Understanding Climate Change, and Chapter 5: Bonding Theories – Explaining Molecular Geometry

Chapter 4: Chemical Bonding – Understanding Climate Change

4.1 Chemical Bonds and Greenhouse Gases
4.2 Electronegativity and Bonding
4.3 Naming Compounds and Writing Formulas
4.4 Lewis Symbols and Lewis Structures
4.5 The Lengths and Strengths of Covalent Bonds
4.6 Resonance
4.7 Formal Charge: Choosing among Lewis Structures (Know how to calculate formal charge and use it in determining the optimal structure of a molecule)
4.8 Exceptions to the Octet Rule
4.9 Vibrating Bonds and the Greenhouse Effect (Not testable, good FYI material)

Read the Summary on page 186
Review the Problem-Solving Summary on page 187

Chapter 5: Bonding Theories – Explaining Molecular Geometry

5.1 Biological Activity and Molecular Shape
5.2 Valence-Shell Electron-Pair Repulsion and Molecular Shape
5.3 Polar Bonds and Polar Molecules
5.4 Valence Bond Theory and Hybrid Orbitals (Know hybrid orbitals and how to determination if a molecule is SP, SP^2 or SP^3)
5.5 Molecules with Multiple "Central" Atoms
5.6 Chirality and Molecular Recognition (Not testable, good FYI material)
5.7 Molecular Orbital Theory (Not testable, good FYI material)

Read the Summary on page 244
Review the Problem-Solving Summary on page 245

Monday 3/1: AP Chem Exam #9, Chapters 5 and 4, and Thermochemistry/Thermodynamics: Chapters 10 and 11

Unit 10 – Intermolecular Forces, Gases and Properties of Solutions
3/2 – 3/14

Non-instructional Days: 3/9 Professional Day, no classes

Read Chapters 6, 9 and 12

Chapter 6: Intermolecular Forces: Attractions between Particles

6.1 Intramolecular Forces versus Intermolecular Forces
6.2 London Dispersion Forces
6.3 Interactions Involving Polar Molecules
6.4 Trends in Solubility
6.5 Phase Diagrams: Intermolecular Forces at Work (You do not need to know the specifics of phase diagrams, but you should have a familiarity with them.)
6.6 Some Remarkable Properties of Water

Read the Summary on page 284
Review the Problem-Solving Summary on page 285

Chapter 9: Properties of Gases

9.1 An Invisible Necessity: The Properties of Gases
9.2 Gas Diffusion and the Kinetic Molecular Theory of Gases (Root Mean Square or Boltzmann constant of the KE of a gas are not testable, nor are the equations on pages 406 – 408; know the correlation between temperature, molar mass and velocity of a gas, but you will not need to know how to calculate these.)
9.3 The Air We Breathe (Understand the working of a manometer.)
9.4 Relating P, T, and V: The Gas Laws (It is NOT important that you know the names of the gas laws named for Boyle, Charles, Avogadro and Amontons. Understand what's written here, but the Laws you really need to know are the Combined Gas Law (next section) and the Ideal Gas Law (section after the next section)
9.5 The Combined Gas Law
9.6 Ideal Gases and the Ideal Gas Law (You should have read 9.6 and 9.7 during semester 1 in the Chapter 7 section of foundations, so this should be simply a quick review.)
9.7 Densities of Gases (See note to section 9.6 which is ditto for 9.7)
9.8 Gases in Chemical Reactions
9.9 Mixtures of Gases (Know how and when to use Dalton's Law of Partial Pressures)
9.10 Real Gases (Largely not testable – read it, but you don't need to know how to use the van der Waals equation, it's not in your reference packet.)

Read the Summary on page 440
Review the Problem-Solving Summary on page 441

Chapter 12: Properties of Solutions: Their Concentrations and Colligative Properties

12.1 Osmosis: “Water, Water, Everywhere” (Review, not testable)
12.2 Osmotic Pressure and the van ’t Hoff Factor (Review, not testable)
12.3 Vapor Pressure
12.4 Solutions of Volatile Substances
12.5 More Colligative Properties of Solutions (Review, not testable)
12.6 Henry’s Law and the Solubility of Gases

Read the Summary on page 588
Review the Problem-Solving Summary on page 588

Monday 3/15: AP Chem Exam #10, Chapters 6, 9 and 12, and Chapters 13 and 17 (Kinetics/Electrochemistry)

Unit 11 – Mass Spectroscopy, PES, Chromatography and IR Analysis
3/16 – 3/17

Read Sections 2.6 (Mass Spec), 3.11 (PES), class notes (Chromatography) and 8.2 (IR analysis, Beers Law)

Thursday 3/18: AP Chem Exam #11,
Sections 2.6 (Mass Spec), 3.11 (PES), class notes (Chromatography) and 8.2 (IR analysis, Beers Law), and Chapters 14, 15, and 16 (Equilibrium and Acid/Base Equilibrium)

This is the last summative assessment grade for the 3rd quarter, which ends on 4/9.

Quarter-4: AP Chemistry Review Schedule

- **Friday 3/19 to Thursday 3/25 – Review practice sets in groups**

3/26 to 4/2 Spring break

- ***Monday 4/5 to Friday 4/9 - Review practice sets in groups***

Monday 4/12: Grading and Planning, no classes

- ***Tuesday 4/13 – Friday 4/16: 4- Practice MC exams (All Task Assessments & Practice Prep)***

AP Practice Test #1:
Monday 4/19: MC
Tuesday 4/20: FRQs
Wednesday 4/21: Grading, Exam discussion

Thursday 4/22: No classes, designated non-instructional day

AP Practice Test #2:
Friday 4/23: MC
Monday 4/26: FRQs
Tuesday 4/27: Grading, Exam discussion

AP Practice Test #3
Wednesday 4/28: MC
Thursday 4/29: FRQs
Friday 4/30: Grading, Exam discussion

PRACTICE:
Monday 5/3: MC
Tuesday 5/4: FRQs
Wednesday 5/5: Grading, Exam discussion

AP Chemistry Exam
May 6 at 12 PM

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